

PHYS 143 – Mechanics and Thermal Physics

The Second Law of Thermodynamics

Why do natural processes run in one direction only, and what limits how much of the heat from a hot source an engine can turn into work?

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Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 20 (§20.1 to §20.7; §20.8 in the supplement)

By the end of this week, you should be able to:

1. Distinguish reversible from irreversible processes, and calculate the entropy change of a system and its surroundings to decide whether a process is allowed.
2. Analyse a heat engine or a refrigerator with an energy-flow diagram, including from a pV cycle.
3. State the second law and use the Carnot cycle to find the maximum efficiency and to test a claimed device.
4. Analyse an unfamiliar engine, refrigerator or entropy problem, choosing the physics without being told.

The detailed outcomes are in the closing checklist.

Three contacts:

1 DIRECTION

Why do processes run only one way, and what is entropy?

2 CONVERT

How do engines and refrigerators move energy between hot and cold?

3 LIMITS

What does the second law forbid, and what is the best an engine can do?

Not marked. Answer individually, then we review.

1. A gas goes round a clockwise cycle, absorbing 900 J in total and rejecting 600 J in total. Find the net work done by the gas and ΔU for the cycle.
2. How much heat warms 0.50 kg of water by 40 K ($c = 4186 \text{ J}/(\text{kg K})$)?
3. How much heat melts 0.10 kg of ice at 0°C ($L_f = 3.34 \times 10^5 \text{ J/kg}$)?
4. Convert 27°C and 327°C to kelvin.
5. Evaluate $\int_{300}^{600} \frac{dT}{T}$.
6. A diatomic gas ($\gamma = 1.40$) is compressed adiabatically to one-eighth of its volume. By what factor does T rise ($TV^{\gamma-1} = \text{constant}$)?

1. $W_{\text{net}} = Q_{\text{net}} = 900 - 600 = 300 \text{ J};$
 $\Delta U = 0$ for a cycle.

2. $Q = mc\Delta T = (0.50)(4186)(40) =$
 $8.4 \times 10^4 \text{ J}.$

3. $Q = mL_f = (0.10)(3.34 \times 10^5) =$
 $3.34 \times 10^4 \text{ J}.$

4. 300 K and 600 K.

5. $\ln(600/300) = \ln 2 = 0.693.$

6. $T_2/T_1 = 8^{0.40} = 2.30.$

The cycle ($\Delta U = 0$), the ledger and the adiabatic relation (Lecture 11) and $Q = mc\Delta T$ and latent heat (Lectures 9 and 10) are all reused. Lecture 11 named the thermal efficiency and left it for this lecture.

Contact 1: Direction

Why do natural processes run only one way, and what is entropy?

An opening puzzle: films run backwards

Poll

Four short films are run backwards. (A) A ball that bounces higher each time. (B) A hot cup of tea that gets hotter in a cool room. (C) Gas that gathers from a whole room into one corner. (D) A pendulum with no friction. **Which could really happen? Which break the first law?**

Predict first

Sketch each final state. For each film, is energy conserved? Is the film believable? Compare your two answers.

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Predict first

Sketch each final state. For each film, is energy conserved? Is the film believable? Compare your two answers.

Only (D) looks natural. Films (A), (B) and (C) conserve energy, *so the first law does not forbid them*. Something else gives processes a direction. That is the second law.

Rule: reversible and irreversible

A process is **reversible** if the system and its surroundings can both be returned exactly to their initial states. Otherwise it is **irreversible**. A reversible process is a series of equilibrium states: quasi-static, with no friction and no finite temperature difference.

Causes of irreversibility

- friction and viscosity (work becomes heat);
- heat flow across a *finite* temperature difference;
- free expansion into a vacuum;
- mixing of different substances.

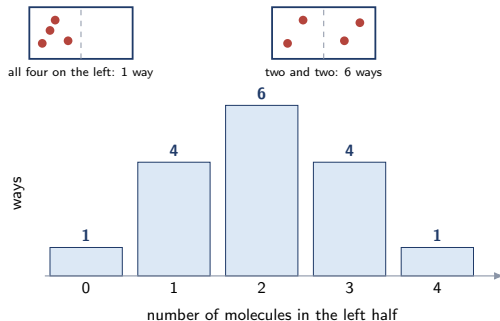
Common error

“If energy is conserved, the process is possible.” The first law does not give a direction. Real processes are irreversible; reversible ones are an idealisation that real processes approach.

TOOLKIT 1 – Reversible or irreversible?

1. **SKETCH** the process and its surroundings, before and after.
 2. **RUN THE FILM BACKWARDS:** would it look natural? If not, the process is irreversible.
 3. **LOOK FOR THE CAUSE:** friction, a finite temperature difference, a sudden expansion, mixing.
 4. **COMPARE** with the idealisation: quasi-static, equilibrium at every step, no friction.
- WATCH OUT:** energy conservation alone cannot decide; and “irreversible” does not mean “impossible”, only “one direction”.
- CHECK:** can both the system and the surroundings be restored without any net effect elsewhere?

Why one direction: counting the ways



- Four molecules in a box: $2^4 = 16$ arrangements; only **1** puts all on the left, **6** give two and two.
- For 100 molecules the chance that all are on the left is $2^{-100} \approx 8 \times 10^{-31}$.

Physical meaning

Entropy is a measure of how many microscopic arrangements correspond to a state. The spread-out state has overwhelmingly more, so it is overwhelmingly more likely. “Disorder” means *number of ways*.

The entropy rule

Rule: entropy change for a reversible isothermal process

$$\Delta S = \frac{Q}{T} \quad [\text{J/K}]$$

Q is the heat added *reversibly* (positive into the system); T is the constant temperature in **kelvin**. In general $dS = dQ_{\text{rev}}/T$.

Physical meaning

S is a **state function**, like U : ΔS depends only on the end states. For an irreversible process, calculate it along a *reversible* path between the same states.

Common errors

(1) Using the heat of an irreversible process. (2) Using $^{\circ}\text{C}$.

TOOLKIT 2 – Entropy change

1. **IDENTIFY** the system and the initial and final states. Is T constant?
 2. **CHOOSE A REVERSIBLE PATH** between the same states.
 3. **SELECT:** constant T (phase change, reservoir): $\Delta S = Q/T$, with $Q = \pm mL$.
Changing T : $\Delta S = mc \ln(T_2/T_1)$.
 4. **PREDICT the sign** first: heat in at a given T gives $\Delta S > 0$; heat out gives $\Delta S < 0$.
- WATCH OUT:** kelvin only; do not use the heat of the irreversible process.
- CHECK:** units J/K; sign agrees with the prediction.

Question

0.200 kg of ice at 0°C melts reversibly to water at 0°C ($L_f = 3.34 \times 10^5 \text{ J/kg}$). Find the entropy change of the ice and water.

Predict first

Sketch the heating curve: a flat section at 273 K. Does the entropy of the system rise or fall? Is the number of ways larger in the liquid or in the solid?

Question

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Predict first

Sketch the heating curve: a flat section at 273 K. Does the entropy of the system rise or fall? Is the number of ways larger in the liquid or in the solid?

Set up: constant $T = 273.15 \text{ K}$; $Q = mL_f = (0.200)(3.34 \times 10^5) = 6.68 \times 10^4 \text{ J}$ into the system.

Execute: $\Delta S = \frac{Q}{T} = \frac{6.68 \times 10^4}{273.15} = 2.45 \times 10^2 \text{ J/K}$.

Evaluate: positive, as predicted: the liquid has more arrangements. Freezing the same water gives $-2.45 \times 10^2 \text{ J/K}$ for the water. ✓

When the temperature changes

Rule: entropy change on heating

$$\Delta S = \int \frac{dQ}{T} = \int_{T_1}^{T_2} \frac{mc dT}{T} = mc \ln \frac{T_2}{T_1}$$

Constant c ; T in kelvin; negative if the body cools.

Physical meaning

Heat the body in many small steps, each at almost constant T : add $dS = mc dT/T$ for each step. The same heat adds more entropy at a low temperature than at a high one.

Common error

Using $\ln(T_2/T_1)$ with Celsius values, or writing $\Delta S = mc\Delta T/T$ with one temperature. Use the logarithm of the *kelvin* ratio.

Your turn

Find the entropy change of 0.50 kg of water heated from 20°C to 80°C ($c = 4186 \text{ J}/(\text{kg K})$).

Predict first

Sketch the heating path. Positive or negative? About 100, 400 or 4000 J/K? ($Q \approx 1.3 \times 10^5 \text{ J}$ near 320 K.)

Your turn

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Predict first

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$$\Delta S = (0.50)(4186) \ln \frac{353.15}{293.15} = 2093 \times 0.1861 = 3.9 \times 10^2 \text{ J/K. Check:}$$
$$Q/T_{\text{av}} \approx 1.26 \times 10^5 / 322 \approx 390 \text{ J/K. } \checkmark$$

Rule: the second law for entropy

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

= 0 for a reversible process; > 0 for an irreversible one; < 0 is **forbidden**.

Physical meaning

The entropy of a *part* can fall (a freezer, a refrigerator) provided the entropy of the surroundings rises by at least as much.

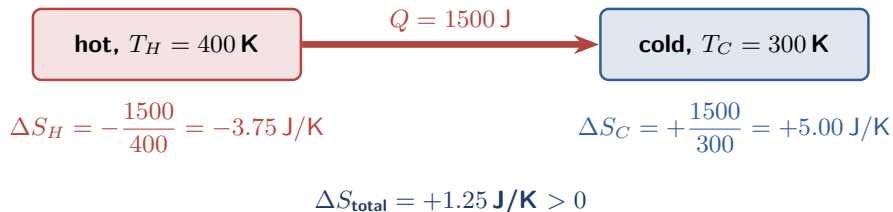
Common error

“Entropy always increases.” Only the *total* for system and surroundings (an isolated whole) cannot decrease.

TOOLKIT 3 – Entropy bookkeeping

1. **LIST** every body that gains or loses heat: system and each reservoir.
 2. **FOR EACH:** $\Delta S_i = Q_i/T_i$ (reservoir, constant T) with Q_i signed for *that* body.
 3. **ADD:** $\Delta S_{\text{total}} = \sum \Delta S_i$.
 4. **DECIDE:** > 0 allowed and irreversible; $= 0$ allowed and reversible; < 0 **impossible**.
- WATCH OUT:** the hot body loses heat ($Q < 0$ for it); the cold body gains ($Q > 0$).
- CHECK:** the cold reservoir's gain exceeds the hot reservoir's loss, since the same $|Q|$ is divided by a smaller T .

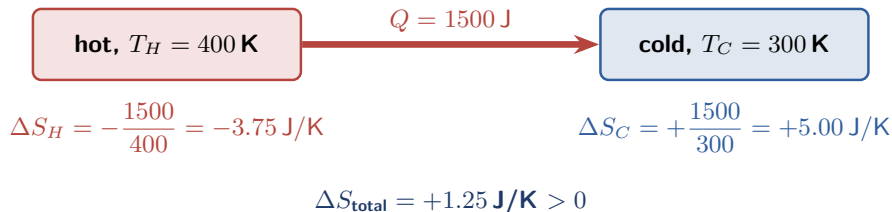
Lecturer example: heat flows from hot to cold



Predict first

A reservoir at 400 K passes 1500 J to one at 300 K. Does the total entropy rise, fall or stay the same? Now try the reverse: 1500 J from cold to hot.

Lecturer example: heat flows from hot to cold

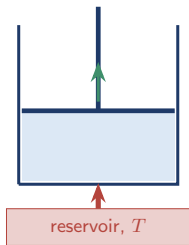


Predict first

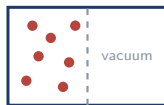
A reservoir at 400 K passes 1500 J to one at 300 K. Does the total entropy rise, fall or stay the same? Now try the reverse: 1500 J from cold to hot.

Forward: $\Delta S_{\text{total}} = -3.75 + 5.00 = +1.25 \text{ J/K} > 0$: allowed. **Reverse:** $\Delta S_{\text{total}} = +3.75 - 5.00 = -1.25 \text{ J/K} < 0$: forbidden. Heat flows hot to cold by itself, never cold to hot (without help).

Lecturer example: expansion, reversible or free



reversible isothermal
expansion



$$Q = 0, W = 0$$

free expansion
(insulated)

0.50 mol of an ideal gas at 300 K expands from 4.0 L to 12.0 L. Find ΔS of the gas and ΔS_{total} in each case.

Predict first

Free expansion has $Q = 0$. Is ΔS of the gas then zero? Which case is irreversible?

Set up: S is a state function, so ΔS_{gas} is the *same* for both. Use the reversible isothermal path: $Q = W = nRT \ln(V_2/V_1)$ (Lecture 11).

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Gas: $\Delta S_{\text{gas}} = \frac{Q}{T} = nR \ln \frac{V_2}{V_1} = (0.50)(8.314) \ln 3 = 4.57 \text{ J/K}$ ($Q = 1.37 \times 10^3 \text{ J}$).

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Reversible: the reservoir gives up $Q = 1.37 \text{ kJ}$ at 300 K : $\Delta S_{\text{surr}} = -4.57 \text{ J/K}$; $\Delta S_{\text{total}} = 0$.

Free expansion: $Q = 0$ for the surroundings: $\Delta S_{\text{surr}} = 0$; $\Delta S_{\text{total}} = +4.57 \text{ J/K} > 0$.

Evaluate: the gas ends in the same state, so its ΔS is equal; only the surroundings differ. The free expansion is irreversible, and its total entropy rises. ✓

Concept check: ranking irreversibility

Choose, then explain

Rank the total entropy change, largest first. (i) A reversible isothermal expansion. (ii) 1500 J flows from 400 K to 300 K. (iii) 1500 J flows from 400 K to 350 K.

A. i, ii, iii B. ii, iii, i C. iii, ii, i D. ii, i, iii E. all equal

Predict first

Sketch the two heat flows. Which has the larger temperature difference? Which is closer to the reversible limit?

Concept check: ranking irreversibility

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Predict first

Sketch the two heat flows. Which has the larger temperature difference? Which is closer to the reversible limit?

(ii) $+1.25$; (iii) $\frac{1500}{350} - \frac{1500}{400} = +0.54$; (i) 0 . Ranking: **B**. The smaller the temperature difference, the closer to reversible: a small difference gives a small ΔS_{total} .

Contact 1: what we can now do

Reversible or irreversible?



Entropy: $\Delta S = Q/T, \quad mc \ln(T_2/T_1)$



Total: $\Delta S_{\text{total}} \geq 0$

You should be able to

- decide whether a process is reversible, and why;
- calculate ΔS for a phase change and for heating;
- find ΔS of system, surroundings and total;
- use ΔS_{total} to decide if a process is allowed.

✓ Before Contact 2

Heat flows from hot to cold by itself, and that flow is energy. Could a machine sit in the flow and take work out of it? Read Young and Freedman, §20.2 to §20.4.

Contact 2: Convert

*How do engines and refrigerators
move energy between hot and cold?*

An opening question: using the flow

Sketch first

Heat flows from a hot reservoir to a cold one. Draw a device between them that produces work. Show every energy arrow and say where all the energy goes.

Predict first

Must all the heat taken from the hot reservoir become work? If the device returns to its starting state after each cycle, what does the first law say about the work?

An opening question: using the flow

Sketch first

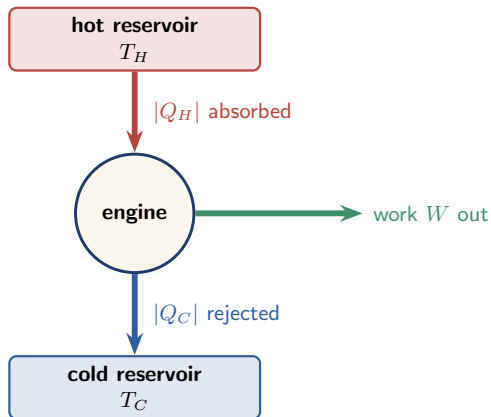
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Predict first

Must all the heat taken from the hot reservoir become work? If the device returns to its starting state after each cycle, what does the first law say about the work?

The device must reject *some* heat to a cold reservoir. We now count the energy: the work out equals the heat in minus the heat rejected.

The heat engine



Rule: heat engine

$$W = |Q_H| - |Q_C|$$

$$e = \frac{W}{|Q_H|} = 1 - \frac{|Q_C|}{|Q_H|}$$

$|Q_H|$, $|Q_C|$: **magnitudes** of the heats per cycle.

- The working substance returns to its start each cycle: $\Delta U = 0$, so $W = Q_{\text{net}}$ (Lecture 11).
- $0 < e < 1$; the energy in is shared between work and rejected heat.

TOOLKIT 4 – Energy flow in an engine

1. **DRAW** the hot and cold reservoirs, the engine and three arrows: $|Q_H|$ in, W out, $|Q_C|$ out.
 2. **USE MAGNITUDES:** $W = |Q_H| - |Q_C|$ for each cycle (first law, $\Delta U = 0$).
 3. **EFFICIENCY:** $e = W/|Q_H|$ (what you get over what you pay). For power, multiply per-cycle quantities by the cycle rate.
 4. **PREDICT:** $|Q_C| > 0$ always, so $e < 1$; $W < |Q_H|$.
- WATCH OUT:** $e = W/|Q_C|$ or $|Q_C|/|Q_H|$ (wrong ratios); mixing signed and magnitude forms of Q .
- CHECK:** $|Q_H| = W + |Q_C|$, and $0 < e < 1$.

Question

In each cycle an engine absorbs 8000 J from a hot reservoir and rejects 5600 J to a cold reservoir. Find (a) the work per cycle, (b) the efficiency, (c) the power at 25 cycles per second.

Predict first

Sketch the three arrows. Will e be more or less than $\frac{1}{2}$? Will the power be about 6, 60 or 600 kW?

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Predict first

Sketch the three arrows. Will e be more or less than $\frac{1}{2}$? Will the power be about 6, 60 or 600 kW?

Execute: (a) $W = |Q_H| - |Q_C| = 8000 - 5600 = 2400 \text{ J}$. (b) $e = W/|Q_H| = 2400/8000 = 0.30$. (c) $P = (2400)(25) = 6.0 \times 10^4 \text{ W} = 60 \text{ kW}$.

Evaluate: $e < \frac{1}{2}$ and about 60 kW, as predicted; 0.30 means that 70% of the heat is rejected.

✓

Your turn

A power station delivers 600 MW of electrical power at an efficiency of 0.36. Find (a) the rate at which heat is supplied to it, (b) the rate at which heat is rejected to the river.

Predict first

Sketch the three arrows with widths in proportion. Is the rejected power more or less than the electrical power?

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(a) $P_H = P/e = 600/0.36 = 1.67 \times 10^3 \text{ MW}$. (b)

$P_C = P_H - P = 1667 - 600 = 1.07 \times 10^3 \text{ MW}$.

Evaluate: the rejected power (1.07 GW) exceeds the electrical power: a 36% plant rejects almost twice what it delivers. ✓

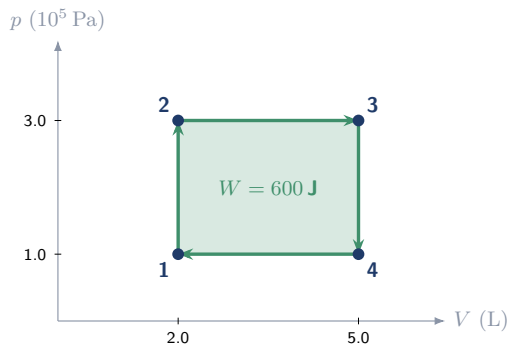
TOOLKIT 5 – Cycle analysis for efficiency

1. **LABEL** the states on the pV diagram; find each T from $pV = nRT$ (ratios are enough).
2. **LEDGER** (Lecture 11): for every leg find W , $\Delta U = nC_V\Delta T$ and $Q = \Delta U + W$.
3. **SORT** the legs: $Q > 0$ is heat *in*; $Q < 0$ is heat *out*. $|Q_H| = \sum Q_{>0}$, $|Q_C| = \sum |Q_{<0}|$.
4. **CLOSE:** $\sum \Delta U = 0$ and $W_{\text{net}} = |Q_H| - |Q_C|$; then $e = W_{\text{net}}/|Q_H|$.

WATCH OUT: an *isochoric* leg that raises T absorbs heat (no work, but $Q > 0$): it belongs in $|Q_H|$.

CHECK: net work equals the enclosed area.

Lecturer example: a rectangular cycle



A monatomic ideal gas ($C_V = \frac{3}{2}R$, $C_p = \frac{5}{2}R$) goes round the cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ shown. Find the net work, the heat absorbed per cycle and the efficiency.

Predict first

Which legs do you expect to take heat in? Is e more or less than 0.5?

Rectangular cycle: solution

leg	W (J)	ΔU (J)	$Q = \Delta U + W$ (J)	rule used
$1 \rightarrow 2$ (isochoric, up)	0	+600	+600	$\Delta U = \frac{3}{2}V\Delta p$
$2 \rightarrow 3$ (isobaric, expand)	+900	+1350	+2250	$W = p\Delta V$
$3 \rightarrow 4$ (isochoric, down)	0	-1500	-1500	$\Delta U = \frac{3}{2}V\Delta p$
$4 \rightarrow 1$ (isobaric, compress)	-300	-450	-750	$W = p\Delta V$
total	+600	0	+600	closes

Rectangular cycle: solution

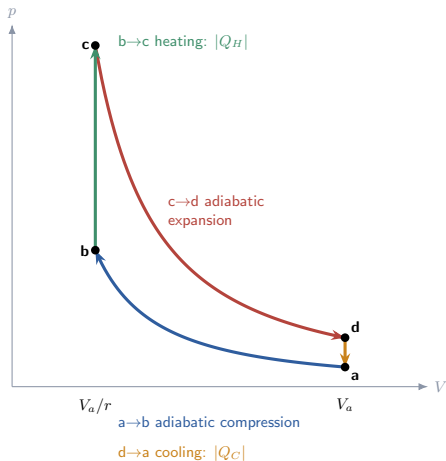
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total	+600	0	+600	closes

Sort: heat *in* on legs $1 \rightarrow 2$ and $2 \rightarrow 3$: $|Q_H| = 600 + 2250 = 2850$ J; heat out on $3 \rightarrow 4$ and $4 \rightarrow 1$: $|Q_C| = 2250$ J.

Result: $e = \frac{W}{|Q_H|} = \frac{600}{2850} = 0.21$.

Evaluate: $\sum \Delta U = 0$ and $W = \text{enclosed area} = (2 \times 10^5)(3 \times 10^{-3}) = 600$ J. ✓ The efficiency is low because the isochoric leg $1 \rightarrow 2$ costs heat but does no work.

The internal-combustion engine: the Otto cycle



Idealised petrol engine, $r = V_a/V_b$ the compression ratio:

1. $a \rightarrow b$ compression stroke (adiabatic);
2. $b \rightarrow c$ ignition (isochoric heating, $|Q_H|$);
3. $c \rightarrow d$ power stroke (adiabatic);
4. $d \rightarrow a$ exhaust (isochoric cooling, $|Q_C|$).

Predict first

Will a larger r raise or lower the efficiency?

Otto cycle: the efficiency

Heats: $|Q_H| = nC_V(T_c - T_b)$, $|Q_C| = nC_V(T_d - T_a)$, so $e = 1 - \frac{T_d - T_a}{T_c - T_b}$.

Otto cycle: the efficiency

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Adiabats ($TV^{\gamma-1}$ constant, Lecture 11): $T_b = T_a r^{\gamma-1}$, $T_d = T_c / r^{\gamma-1}$.

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Adiabats ($TV^{\gamma-1}$ constant, Lecture 11): $T_b = T_a r^{\gamma-1}$, $T_d = T_c / r^{\gamma-1}$.

Combine: $T_d - T_a = \frac{T_c}{r^{\gamma-1}} - \frac{T_b}{r^{\gamma-1}} = \frac{T_c - T_b}{r^{\gamma-1}}$.

Rule: efficiency of the Otto cycle

$$e = 1 - \frac{1}{r^{\gamma-1}}$$

Ideal gas with constant C_V , reversible adiabats; it depends only on r and γ .

Question

0.020 mol of air ($C_V = \frac{5}{2}R$, $\gamma = 1.40$) is at 300 K at the start of compression in an Otto cycle with $r = 8.0$. Ignition adds $|Q_H| = 500$ J. Find the temperatures T_b , T_c , T_d , the heat rejected, the work and the efficiency.

Predict first

Sketch the cycle. Will e be about 0.3, 0.55 or 0.9? Which of T_b , T_c is the largest temperature?

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Predict first

Sketch the cycle. Will e be about 0.3, 0.55 or 0.9? Which of T_b , T_c is the largest temperature?

Execute: $nC_V = (0.020)(20.79) = 0.416$ J/K. $T_b = 300(8.0)^{0.40} = 689$ K.
 $T_c = T_b + |Q_H|/(nC_V) = 689 + 1203 = 1892$ K. $T_d = T_c/8.0^{0.40} = 824$ K.
 $|Q_C| = nC_V(T_d - T_a) = (0.416)(524) = 218$ J; $W = 500 - 218 = 282$ J; $e = 282/500 = 0.56$.
Evaluate: formula $e = 1 - 8.0^{-0.40} = 0.565$. ✓ Real petrol engines reach about 0.25 to 0.35: friction, heat losses and non-adiabatic strokes.

Concept check: the compression ratio

Choose, then explain

Raising the compression ratio from 8 to 12 for $\gamma = 1.40$:

- A. e falls B. e rises from 0.565 to 0.630 C. e rises to 0.90 D. e is unchanged
E. e doubles

Predict first

Sketch e against r : rising steeply at first and then flattening? Does e reach 1 for any r ?

Choose, then explain

Raising the compression ratio from 8 to 12 for $\gamma = 1.40$:

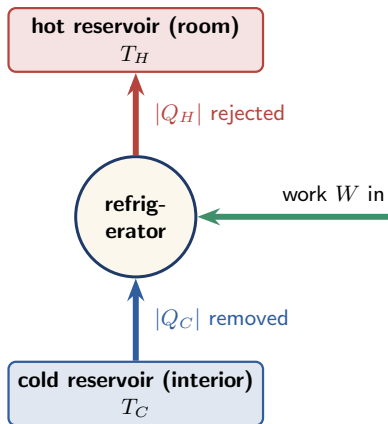
- A. e falls B. e rises from 0.565 to 0.630 C. e rises to 0.90 D. e is unchanged
E. e doubles

Predict first

Sketch e against r : rising steeply at first and then flattening? Does e reach 1 for any r ?

$e = 1 - 12^{-0.40} = 1 - 0.370 = 0.630$: **B**. The curve flattens; $e < 1$ for every finite r . In a real petrol engine a high r ignites the fuel too early (knocking), which limits r to about 10.

The refrigerator



Rule: refrigerator

$$|Q_H| = |Q_C| + W$$

$$K = \frac{|Q_C|}{W}$$

K : coefficient of performance.

- A heat engine run backwards: work W in moves heat from cold to hot.
- K can exceed 1. A good K is large.

TOOLKIT 6 – Refrigerators and heat pumps

1. **DRAW** the same diagram as the engine with the arrows reversed: $|Q_C|$ in from the cold space, W in, $|Q_H|$ out to the hot space.
 2. **ENERGY:** $|Q_H| = |Q_C| + W$ (first law for a cycle).
 3. **PERFORMANCE:** refrigerator $K = |Q_C|/W$; heat pump (heating the hot space) $K_{\text{hp}} = |Q_H|/W$.
 4. **PREDICT:** $|Q_H| > |Q_C|$: the hot space receives more heat than the cold space loses.
- WATCH OUT:** K is not an efficiency, and it is not limited to 1.
- CHECK:** $|Q_H| - |Q_C| = W > 0$.

Question

In each cycle a refrigerator removes 600 J from its interior using 200 J of electrical work. Find K and the heat released into the kitchen.

Predict first

Sketch the arrows. Is the heat rejected to the kitchen more than, equal to, or less than 600 J?

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$$K = \frac{|Q_C|}{W} = \frac{600}{200} = 3.0. \quad |Q_H| = |Q_C| + W = 600 + 200 = 800 \text{ J.}$$

Evaluate: $800 > 600$, as predicted: the kitchen gains 800 J per cycle. ✓

Your turn

An air conditioner with $K = 2.5$ removes heat from a room at the rate of 5.0 kW . Find the electrical power it uses and the rate at which it releases heat outdoors.

Predict first

Sketch the three arrows. Is the electrical power more or less than 5.0 kW ? Is the outdoor heat rate more or less than 5.0 kW ?

Your turn

An air conditioner with $K = 2.5$ removes heat from a room at the rate of 5.0 kW. Find the electrical power it uses and the rate at which it releases heat outdoors.

Predict first

Sketch the three arrows. Is the electrical power more or less than 5.0 kW? Is the outdoor heat rate more or less than 5.0 kW?

$$P = \frac{P_C}{K} = \frac{5.0}{2.5} = 2.0 \text{ kW}; \quad P_H = P_C + P = 7.0 \text{ kW}.$$

Evaluate: 2.0 kW of electricity moves 5.0 kW of heat: possible because $K > 1$ (the work only moves the heat). ✓

Concept check: the open refrigerator door

Choose, then explain

A refrigerator is left open in a closed, insulated kitchen. The kitchen:

A. cools B. stays at the same temperature C. warms D. depends on K

Predict first

Sketch the kitchen as one system with electricity entering. What is the net energy flow across the walls?

Concept check: the open refrigerator door

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A. cools B. stays at the same temperature C. warms D. depends on K

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C. The heat removed from the air is released back into the same room, plus the work: the net effect is $|Q_H| - |Q_C| = W > 0$. The kitchen warms by the electrical energy used. A refrigerator cools only if its hot side is outside the room.

Contact 2: what we can now do

Engine: $W = |Q_H| - |Q_C|$, $e = W/|Q_H|$



pV cycle: heat in and out leg by leg; Otto
 $1 - 1/r^{\gamma-1}$



Refrigerator: $|Q_H| = |Q_C| + W$,
 $K = |Q_C|/W$

You should be able to

- draw the energy-flow diagram and use magnitudes;
- find e from the heats or from a pV cycle;
- apply the Otto efficiency;
- use K for refrigerators and heat pumps.

✓ Before Contact 3

Every engine rejects heat: a flaw, or a law of nature? What is the best e between two temperatures? Read Young and Freedman, §20.5 to §20.7.

Contact 3: Limits

*What does the second law forbid, and
what is the best an engine can do?*

An opening question: the ship

Sketch first

A ship's engine takes heat from the ocean at 290 K and turns *all* of it into work to drive the propeller. The ocean holds far more energy than the ship will ever use. Does this violate the first law? Why can it not be built?

Predict first

Sketch the energy-flow diagram. How many reservoirs does the device use? What is $|Q_C|$?

An opening question: the ship

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Predict first

Sketch the energy-flow diagram. How many reservoirs does the device use? What is $|Q_C|$?

It does *not* violate the first law ($W = |Q_H|$ conserves energy). It uses one reservoir and has $|Q_C| = 0$, so $e = 1$. The second law forbids exactly this.

The second law: three statements

Rule: the second law of thermodynamics

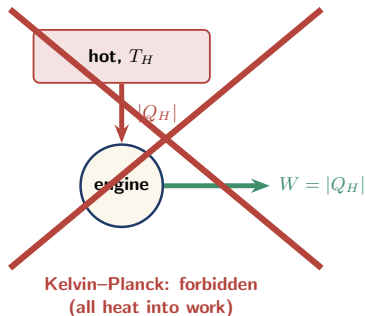
Kelvin–Planck: No cyclic engine can take heat from one reservoir and convert *all* of it into work.

Clausius: No cyclic device can move heat from a cold body to a hot body with *no* work input.

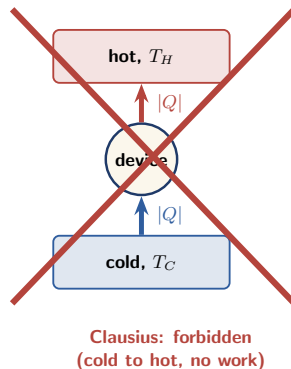
Entropy: $\Delta S_{\text{total}} \geq 0$ for every process; $= 0$ only for a reversible one.

The three are equivalent: a violation of one allows a violation of the others.

The two forbidden devices



Kelvin-Planck: $\Delta S_{\text{total}} = -|Q_H|/T_H < 0$.



Clausius: $\Delta S_{\text{total}} = \frac{|Q|}{T_H} - \frac{|Q|}{T_C} < 0$ for $T_H > T_C$.

The best engine: Carnot's theorem

Rule: Carnot's theorem

No engine operating between two given reservoirs can be more efficient than a **reversible** engine operating between the same reservoirs. All reversible engines between the same reservoirs have the **same** efficiency, whatever the working substance.

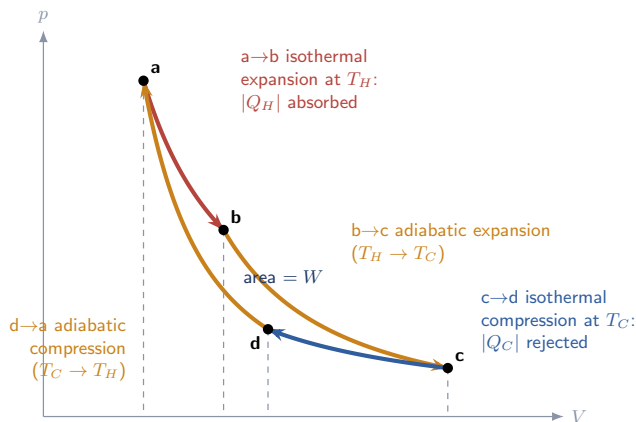
Physical meaning

The limit is set by nature, not by materials or design. It depends only on the two temperatures.

Common error

“A better working substance gives a better limit.” It does not: every reversible engine between T_H and T_C has the same e .

The Carnot cycle



A reversible engine with an ideal gas: two isotherms and two adiabats.

Predict first

Which curves are steeper, the isotherms or the adiabats? At which stages is heat exchanged?

Heat flows only on the isotherms (at T_H in, at T_C out); the adiabats change the temperature with $Q = 0$.

The Carnot cycle: the efficiency

Isotherms (Lecture 11, $Q = W = nRT \ln(V_f/V_i)$): $|Q_H| = nRT_H \ln \frac{V_b}{V_a}$,

$$|Q_C| = nRT_C \ln \frac{V_c}{V_d}.$$

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Adiabats ($TV^{\gamma-1}$ constant): $T_H V_b^{\gamma-1} = T_C V_c^{\gamma-1}$ and $T_H V_a^{\gamma-1} = T_C V_d^{\gamma-1}$.

Dividing: $V_b/V_a = V_c/V_d$, so the two logarithms are *equal*.

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Dividing: $V_b/V_a = V_c/V_d$, so the two logarithms are *equal*.

Ratio: $\frac{|Q_C|}{|Q_H|} = \frac{T_C}{T_H}$. Hence

Rule: Carnot efficiency and refrigerator

$$e_C = 1 - \frac{T_C}{T_H}, \quad K_C = \frac{T_C}{T_H - T_C}$$

T in **kelvin**. The refrigerator form follows from $K = |Q_C|/(|Q_H| - |Q_C|)$.

TOOLKIT 7 – The Carnot limit

1. **IDENTIFY** the two reservoirs and convert both to kelvin.
2. **LIMIT:** $e_{\max} = 1 - T_C/T_H$; for a refrigerator $K_{\max} = T_C/(T_H - T_C)$.
3. **TEST a claim:** compare the claimed e (or K) with the limit; or compute $\Delta S_{\text{total}} = |Q_C|/T_C - |Q_H|/T_H$ per cycle.
4. **DECIDE:** $e > e_C$ or $\Delta S_{\text{total}} < 0$: impossible. $e = e_C$: reversible. $e < e_C$: allowed, irreversible.

WATCH OUT: °C in the formula; treating e_C as a design target (reversible means infinitely slow, with zero power).

CHECK: $0 < e < e_C < 1$.

Entropy and the Carnot limit

Per cycle the engine returns to its start ($\Delta S_{\text{engine}} = 0$). The reservoirs change by

$$\Delta S_{\text{total}} = \frac{|Q_C|}{T_C} - \frac{|Q_H|}{T_H}.$$

Reversible ($|Q_C|/|Q_H| = T_C/T_H$):
 $\Delta S_{\text{total}} = 0$.

Irreversible: $\Delta S_{\text{total}} > 0$, so

$$\frac{|Q_C|}{|Q_H|} > \frac{T_C}{T_H} \Rightarrow e < 1 - \frac{T_C}{T_H}.$$

Physical meaning

The second law in terms of entropy gives the efficiency limit directly. Any irreversibility generates entropy and lowers the work obtained.

One line

$e \leq e_C$ and $\Delta S_{\text{total}} \geq 0$ say the same thing.

Question

An inventor claims an engine operating between reservoirs at 600 K and 300 K that absorbs 4000 J and delivers 2300 J of work per cycle. Is this possible?

Predict first

Sketch the arrows. Is $e = 2300/4000$ more or less than $\frac{1}{2}$? What would be the largest possible e for these reservoirs?

Question

An inventor claims an engine operating between reservoirs at 600 K and 300 K that absorbs 4000 J and delivers 2300 J of work per cycle. Is this possible?

Predict first

Sketch the arrows. Is $e = 2300/4000$ more or less than $\frac{1}{2}$? What would be the largest possible e for these reservoirs?

Limit: $e_C = 1 - 300/600 = 0.50$. **Claim:** $e = 2300/4000 = 0.575 > 0.50$.

Check by entropy: $|Q_C| = 4000 - 2300 = 1700$ J;

$$\Delta S_{\text{total}} = \frac{1700}{300} - \frac{4000}{600} = 5.67 - 6.67 = -1.0 \text{ J/K} < 0.$$

Decision: impossible: it exceeds the Carnot limit and lowers the total entropy. At most 2000 J of work could be obtained.

Power station: steam at 500°C , river at 20°C .

$$T_H = 773.15 \text{ K}, T_C = 293.15 \text{ K:}$$

$$e_C = 1 - \frac{293.15}{773.15} = 0.621.$$

Real stations reach about 0.35 to 0.45: friction, finite temperature differences and heat losses make the cycle irreversible.

Freezer: interior -18°C , kitchen 25°C .

$$T_C = 255.15 \text{ K}, T_H = 298.15 \text{ K:}$$

$$K_C = \frac{255.15}{43.0} = 5.93.$$

To remove 200 W of heat the work input is at least $200/5.93 = 34 \text{ W}$. Real freezers have $K \approx 2$.

Predict first

Both limits are upper bounds. Is each actual device above or below its limit?

Concept check: where to improve a Carnot engine

Choose, then explain

A Carnot engine works between $T_H = 600$ K and $T_C = 300$ K. Which change raises e more? (i) Raise T_H by 50 K. (ii) Lower T_C by 50 K.

A. (i) B. (ii) C. both equal D. neither changes e

Predict first

Write $e = 1 - T_C/T_H$. Which variable changes the ratio more for the same 50 K? Estimate before calculating.

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Choose, then explain

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Predict first

Write $e = 1 - T_C/T_H$. Which variable changes the ratio more for the same 50 K? Estimate before calculating.

Base: $e = 0.500$. (i) $1 - 300/650 = 0.538$. (ii) $1 - 250/600 = 0.583$. **B**: lowering T_C by ΔT changes the ratio by $\Delta T/T_H$, while raising T_H changes it only by $T_C\Delta T/T_H^2$, which is smaller when $T_C < T_H$.

Your turn: allowed or not?

Your turn

An engine absorbs 1500 J from a reservoir at 500 K and rejects 1000 J to a reservoir at 300 K in each cycle. Decide whether it is possible, whether it is reversible, and find ΔS_{total} per cycle.

Predict first

Sketch the arrows. Compare e with $e_C = 1 - 300/500$ before the entropy check.

Your turn: allowed or not?

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Predict first

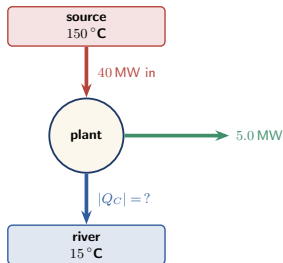
Sketch the arrows. Compare e with $e_C = 1 - 300/500$ before the entropy check.

$$W = 1500 - 1000 = 500 \text{ J}, e = 0.333 < e_C = 0.400.$$

$$\Delta S_{\text{total}} = \frac{1000}{300} - \frac{1500}{500} = 3.33 - 3.00 = +0.33 \text{ J/K} > 0.$$

Evaluate: allowed but irreversible, since $e < e_C$ and $\Delta S_{\text{total}} > 0$ agree. ✓

Integrated problem: decide what physics applies



Decide what physics applies; do not look for a formula

A geothermal plant takes heat from hot water at 150°C and rejects it to a river at 15°C. It receives heat at 40 MW and delivers 5.0 MW of electrical power.

- (a) What is the maximum power any engine could deliver from this heat?
- (b) Find the plant's efficiency and the rate at which it rejects heat.
- (c) Find the rate at which entropy is generated.
- (d) How much of the available power is lost to irreversibility?

Predict first

Sketch the energy flow. Is the actual e above or below the maximum? Is the lost power more or less than half the maximum?

Identify, set up: two reservoirs (Toolkit 7), energy flow (Toolkit 4) and entropy (Toolkit 3).
 $T_H = 423.15 \text{ K}$, $T_C = 288.15 \text{ K}$.

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$$T_H = 423.15 \text{ K}, T_C = 288.15 \text{ K}.$$

$$(a) \quad e_C = 1 - \frac{288.15}{423.15} = 0.319; \quad P_{\max} = e_C P_H = (0.319)(40) = 12.8 \text{ MW}.$$

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$$\text{(b)} \quad e = \frac{5.0}{40} = 0.125; \quad P_C = 40 - 5.0 = 35 \text{ MW}.$$

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$$(c) \dot{S}_{\text{gen}} = \frac{P_C}{T_C} - \frac{P_H}{T_H} = \frac{35 \times 10^6}{288.15} - \frac{40 \times 10^6}{423.15} = 1.215 \times 10^5 - 0.945 \times 10^5 = 2.7 \times 10^4 \text{ W/K}.$$

Identify, set up: two reservoirs (Toolkit 7), energy flow (Toolkit 4) and entropy (Toolkit 3).

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$$(d) T_C \dot{S}_{\text{gen}} = (288.15)(2.69 \times 10^4) = 7.8 \text{ MW}.$$

Evaluate: $e = 0.125 < e_C = 0.319$ (allowed); $\dot{S}_{\text{gen}} > 0$; and

$P_{\max} - P = 12.8 - 5.0 = 7.8 \text{ MW}$, equal to $T_C \dot{S}_{\text{gen}}$: the lost work is T_C times the entropy generated. About 61% of the available power is lost. ✓

Extension (unassessed): the heat pump

A heat pump is a refrigerator whose *hot* side is the house. Its performance is

$$K_{\text{hp}} = \frac{|Q_H|}{W} \leq \frac{T_H}{T_H - T_C}.$$

For a house at 22°C and outside air at 2°C :

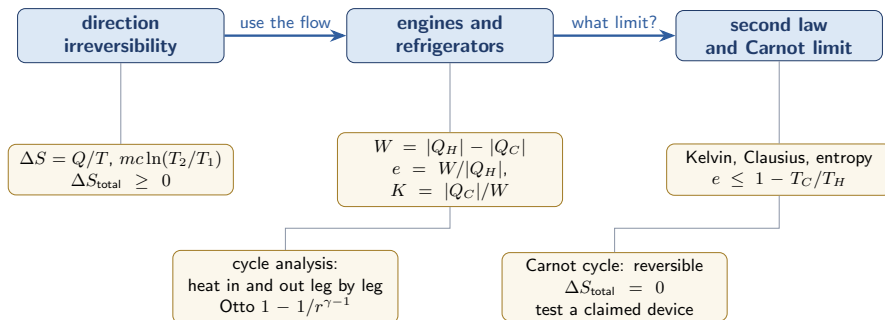
$$K_{\text{hp,max}} = \frac{295.15}{20.0} = 14.8.$$

An application of Toolkits 6 and 7; it is not assessed. Real heat pumps reach about 3 to 4.

Why it beats a heater

An electric heater turns 1 J of work into 1 J of heat. A heat pump with $K_{\text{hp}} = 3$ delivers 3 J of heat for 1 J of work, taking the rest from the outside air.

Lecture 12 concept map



MASTER PROCEDURE – SECOND LAW

1. **IDENTIFY** The system, the reservoirs and the process: heat flow, engine, refrigerator or expansion.
 2. **SKETCH** The energy-flow diagram or the pV cycle, with the arrows.
 3. **UNITS AND SIGNS** Kelvin; magnitudes of Q_H and Q_C ; signed Q in a ledger.
 4. **PREDICT** Allowed or not; the sign of ΔS ; bounds on e or K .
 5. **SELECT THE MODEL** Entropy rule, energy flow, cycle ledger, Otto or Carnot limit.
 6. **EXECUTE** Write the equations first, then substitute with units.
 7. **EVALUATE** Units, $W = |Q_H| - |Q_C|$, $0 < e < e_C$, and $\Delta S_{\text{total}} \geq 0$.
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

The thermal half of the course is now complete: temperature, gases and phases, energy conservation, and the direction and limits of energy conversion.



THE SEVEN TOOLKITS

- | | | |
|---|--------------------------------------|---|
| 1 | Reversible or irreversible? | the film test; causes; first law gives no direction |
| 2 | Entropy change | $\Delta S = Q/T$; $\pm mL/T$; $mc \ln(T_2/T_1)$; kelvin |
| 3 | Entropy bookkeeping | system, surroundings, total; $\Delta S_{\text{total}} \geq 0$ |
| 4 | Energy flow in an engine | $W = Q_H - Q_C $; $e = W/ Q_H $; magnitudes |
| 5 | Cycle analysis for efficiency | ledger; sort legs into heat in and heat out; Otto |
| 6 | Refrigerators and heat pumps | $ Q_H = Q_C + W$; $K = Q_C /W$ |
| 7 | The Carnot limit | $e_C = 1 - T_C/T_H$; K_C ; test a claim |

The Master Procedure sits above these. Earlier toolkits (units, signs, the ledger, adiabatic scaling, $Q = mc\Delta T$) are reused, not renamed.

Entropy

$$\Delta S = Q/T \text{ (reversible, isothermal); } \pm mL/T$$

$$\Delta S = mc \ln(T_2/T_1)$$

$$\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$$

Heat engine

$$W = |Q_H| - |Q_C|,$$

$$e = W/|Q_H| = 1 - |Q_C|/|Q_H|$$

Otto cycle

$$e = 1 - 1/r^{\gamma-1}$$

Refrigerator and heat pump

$$|Q_H| = |Q_C| + W, \quad K = |Q_C|/W,$$

$$K_{\text{hp}} = |Q_H|/W$$

Carnot

$$|Q_C|/|Q_H| = T_C/T_H$$

$$e_C = 1 - T_C/T_H, \quad K_C = T_C/(T_H - T_C)$$

Irreversibility

$$\Delta S_{\text{gen}} = |Q_C|/T_C - |Q_H|/T_H \geq 0; \quad \text{lost}$$

$$\text{work} = T_C \Delta S_{\text{gen}}$$

Six habits for Lecture 12

✓ Habit 1

Run the film backwards: energy conservation does not give the direction.

✓ Habit 2

Use kelvin, and a reversible path, for every entropy calculation.

✓ Habit 3

Add the entropy of the surroundings before you decide.

✓ Habit 4

Draw the energy-flow diagram and use magnitudes: $W = |Q_H| - |Q_C|$.

✓ Habit 5

On a pV cycle, sort every leg into heat in or heat out; an isochoric leg can absorb heat.

✓ Habit 6

Compare any claim with e_C or ΔS_{total} .

Checklist: can I ...

- ☐ distinguish reversible from irreversible and name the causes?
- ☐ calculate $\Delta S = Q/T$ and $\Delta S = mc \ln(T_2/T_1)$?
- ☐ find ΔS of system, surroundings and total?
- ☐ decide from ΔS_{total} whether a process is allowed?
- ☐ draw an energy-flow diagram and use $W = |Q_H| - |Q_C|$?
- ☐ calculate $e = W/|Q_H|$ and engine power?
- ☐ find the heat in and out on each leg of a pV cycle?
- ☐ derive and use the Otto efficiency?
- ☐ use $K = |Q_C|/W$ for a refrigerator and a heat pump?
- ☐ state the three forms of the second law?
- ☐ describe the four stages of the Carnot cycle?
- ☐ use $e_C = 1 - T_C/T_H$ and K_C ?
- ☐ test a claimed engine with e_C and ΔS_{total} ?
- ☐ select the right physics in an unfamiliar problem?

After this week

Tutorial: entropy, mixing, engine, Otto, refrigerator, Carnot and combined problems. Supplement: equivalence of the statements, entropy of an ideal gas, $S = k \ln w$, the Diesel cycle, the heat pump, the arrow of time.